Worksheet 4
One Big Example

Quantum States • Probabilities • Measurement

As a quantum physicist, your job is to prepare an ensemble of spin-1/2 particles, all in the same state given by

$$|\psi\rangle = |+\rangle - 2i|-\rangle.$$

You then make a series of measurements on the ensemble, each time returning it to this original state.

1. Would you say that the particle in your ensemble are more "spin-up" or more "spin-down"? Why?

2. Normalize the state. Why do we need to work with states that are normalized?

3. Write the state as a bra, and write both the ket and the bra in matrix notation.

4. While a particle is in the state $|\psi\rangle$, what is its spin component along the z axis, S_z?

5. What are the possibilities and probabilities when measuring

   (a) S_z?

   (b) S_x?

   (c) S_y?

6. If a measurement of S^2 is made, what will be measured and with what probability?

7. Calculate the expectation values
   (a) $\langle S_z \rangle$ and $\langle S_z^2 \rangle$.

   (b) $\langle S_x \rangle$ and $\langle S_x^2 \rangle$.

(c) $\langle S_y \rangle$ and $\langle S_y^2 \rangle$.

8. Calculate σ_{S_x}, σ_{S_y}, and σ_{S_z}, and show that the uncertainty principle

$$\sigma_{S_x}\sigma_{S_y} \geq \frac{\hbar}{2}|\langle S_z \rangle|$$

holds for this state.